

Hints & Solutions

#MathBoleTohMathonGo

Q1 (3) - Single Correct

We have, $y(x + y^3) dx = x(y^3 - x) dy$

$$\Rightarrow xy dx + y^4 dx = xy^3 dy - x^2 dy$$

$$\Rightarrow xy dx + x^2 dy = xy^3 dy - y^4 dx$$

$$\Rightarrow x(y dx + x dy) = x^2 y^3 \left[\frac{1}{x} dy - \frac{y}{x^2} dx \right]$$

$$\Rightarrow \frac{dxy}{x^2 y^2} = \frac{y}{x} \left[\frac{1}{x} dy - \frac{y}{x^2} dx \right]$$

$$\frac{d(xy)}{(xy)^2} = \frac{y}{x} d\left(\frac{y}{x}\right)$$

On integrating both sides, we get $-\frac{1}{xy} = \frac{1}{2} \left(\frac{y}{x}\right)^2 + C$

It passes through $(4, -2)$

$$\Rightarrow \frac{1}{8} = \frac{1}{2} \left(\frac{-1}{2}\right)^2 + C \Rightarrow C = 0$$

$$\Rightarrow \frac{-1}{xy} = \frac{y^2}{2x^2} \Rightarrow y^3 = -2x$$

$$\therefore f(x) = (-2x)$$

The second equation is given as

$$Y = \int_{1/8}^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_{1/8}^{\cos^2 x} \cos^{-1} \sqrt{t} dt$$

On differentiating w.r.t. x , we get

$$\frac{dy}{dx} = \sin^{-1} \sqrt{\sin^2 x} \cdot 2 \sin x \cos x - \cos^{-1} \sqrt{\cos^2 x} \cdot 2 \sin x \cos x$$

$$= 2 \times \sin x \cos x - 2x \sin x \cos x$$

$$\Rightarrow \frac{dy}{dx} = 0 \Rightarrow y = C$$

Here, put $\sin x = \cos x = \frac{1}{\sqrt{2}}$

$$\Rightarrow y = \int_{1/8}^{1/2} \sin^{-1} \sqrt{t} dt + \int_{1/8}^{1/2} \cos^{-1} \sqrt{t} dt$$

$$y = \int_{1/8}^{1/2} (\sin^{-1} \sqrt{t} + \cos^{-1} \sqrt{t}) dt = \left[\frac{\pi}{2} t \right]_{1/8}^{1/2} = \frac{3\pi}{16}$$

Now area between, $y = (-2x)^{1/3}$ and $y = 3\pi/16$ is

$$\text{At } y = \frac{3\pi}{16}, \quad x = \frac{-1}{2} \left(\frac{3\pi}{16}\right)^3$$

$$\therefore \text{Area} = \int_{-1/2 \left(\frac{3\pi}{16}\right)^3}^0 \frac{1}{2} \left(\frac{3\pi}{16}\right)^3 \left[\frac{3\pi}{16} + (2x)^{1/3} \right] dx$$

$$= \left[\frac{3\pi}{16} x + \frac{3}{4} \cdot 2^{1/3} (x)^{4/3} \right]_{-1/2 \left(\frac{3\pi}{16}\right)^3}^0$$

$$= \frac{1}{8} \left(\frac{3\pi}{16}\right)^4$$

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Q2 (2) - Single Correct

$$\text{Given, } \lim_{t \rightarrow a} \frac{\int_a^t f(x) dx - \left(\frac{t-a}{2}\right) \{f(t) + f(a)\}}{(t-a)^3} = 3a$$

$$\Rightarrow \lim_{t \rightarrow a} \frac{f(t) - \left(\frac{t-a}{2}\right) \left(f'(t) - \frac{1}{2}[f(t) + f(a)]\right)}{3(t-a)^2} = 3a$$

$$\Rightarrow \lim_{t \rightarrow a} \frac{\frac{1}{2}[f(t) - f(a)] - \frac{1}{2}f'(t)(t-a)}{3(t-a)^2} = 3a$$

$$\Rightarrow \lim_{t \rightarrow a} \frac{f'(t) - f''(t)(t-a) - f'(t)}{12(t-a)} = 3a$$

$$\Rightarrow \lim_{t \rightarrow a} -f''(t) = 36a$$

$$\Rightarrow f''(a) = -36a$$

On integrating both sides, we get

$$f'(a) = -18a^2 + C$$

$$\Rightarrow f'(1) = -18 + C$$

$$\Rightarrow 42 = -18 + C \Rightarrow C = 60$$

$$\therefore f'(a) = -18a^2 + 60$$

Again, integrating both sides, we get

$$f(a) = -6a^3 + 60a + C$$

$$\Rightarrow f(1) = -6 + 60 + C$$

$$\Rightarrow -6 = -6 + 60 + C$$

$$\Rightarrow C = -60$$

$$\therefore f(0) = -60$$

Q3 (3) - Single Correct

At the point of maxima

$$f'(x) = 0 \text{ and } f''(x) < 0.$$

$$\Rightarrow f''(x) = -9 + x^2 + f^2(x) < 0$$

$$\Rightarrow x^2 + f^2(x) < 9$$

$$\Rightarrow P[x_1, f(x)] \text{ lies inside the circle } x^2 + y^2 = 9.$$

$\therefore$ No tangent is possible.

Q4 (3) - Single Correct

It is given that, $\frac{dy}{dt} = -k\sqrt{y}$

$$\frac{dy}{\sqrt{y}} = -k dt$$

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On integrating both sides, we get

$$2\sqrt{y} = -kt + C$$

when, $t = 0$ and $y = 4$, then $C = 4$

$$\text{So, } 2\sqrt{y} = -\frac{1}{15}t + 4 \quad [\because k = 1/15]$$

when $y = 0$, then $t = 60$

So, $t = 60$ min

Q5 (1) - Single Correct

We have, $Y = Ax + Be^{2x}$... (i)

On differentiating w.r.t. x , we get

$$\frac{dy}{dx} = A + 2Be^{2x}$$

From Eq. (i), we have

$$\frac{dy}{dx} = A + 2(y - Ax)$$

$$\Rightarrow \frac{dy}{dx} = A + 2y - 2Ax \quad \dots (ii)$$

Again, differentiating w.r.t. x , we get

From Eq. (ii),

$$\frac{d^2y}{dx^2} = \frac{2dy}{dx} - 2A$$

$$\frac{d^2y}{dx^2} = \frac{2dy}{dx} - 2\left(\frac{dy}{dx} - 2y\right)$$

$$\Rightarrow (1 - 2x) \left[\frac{d^2y}{dx^2} - \frac{2dy}{dx} \right] + 2 \left(\frac{dy}{dx} - 2y \right) = 0$$

$$\therefore k = 2 \text{ and } I = -2$$

Q6 (1) - Single Correct

Given, $f'(x) \sin x + f(x) \cos x = 1$

$$\Rightarrow df(x) \sin x = 1$$

On integrating both sides, we get

$$f(x) \sin x = x + C$$

$$\Rightarrow f(x) = \frac{x + C}{\sin x}$$

$f(x)$ being bounded when $x \rightarrow 0$, $C = 0$

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$$f(x) = \frac{x}{\sin x}$$

Now, $\frac{x}{\sin x}$ is increasing function from 0 to $\frac{\pi}{2}$.

$\therefore$ Minimum value of $f(x) = 1$ at $x = 0$

and maximum value of $f(x) = \frac{\pi}{2}$ at $x = \frac{\pi}{2}$

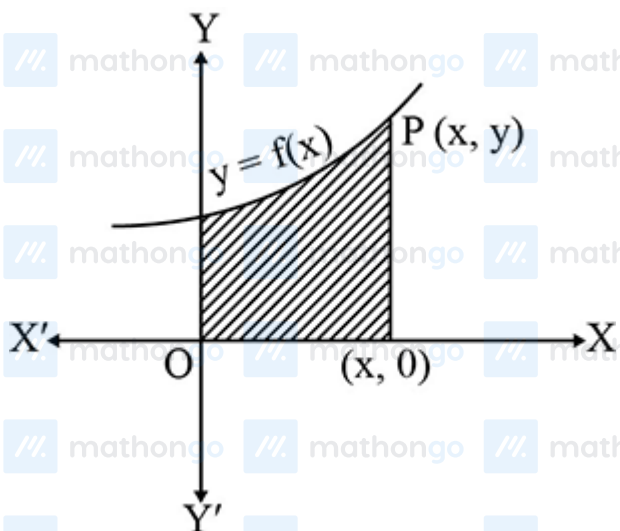
$$\therefore 1 \left(\frac{\pi}{2} - 0 \right) < \int_0^{\pi/2} f(x) dx < \frac{\pi}{2} \left(\frac{\pi}{2} - 0 \right)$$

$$\Rightarrow \frac{\pi}{2} < \int_0^{\pi/2} f(x) dx < \frac{\pi^2}{4}$$

$$\Rightarrow \frac{\pi}{2} < 1 < \frac{\pi^2}{4}$$

Q7 (3) - Single Correct

It is given that, Area of shaded region = y^3



$$\int_0^x y dx = y^3$$

On differentiating w.r.t. x , we get

$$y = 3y^2 \frac{dy}{dx}$$

$$\Rightarrow 3y dy = dx$$

On integrating, both sides, we get

$$\Rightarrow \frac{3y^2}{2} = x + C$$

$\therefore 3y^2 = 2x + 2C$, which represents a parabola.

Q8 (3) - Single Correct

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Given, $y = u^m$

On differentiating w.r.t. x , we get

$$\frac{dy}{dx} = mu^{m-1} \frac{du}{dx}$$

On substituting the values of y and $\frac{dy}{dx}$ in

$$2x^4 y \frac{dy}{dx} + y^4 = 4x^6$$

$$\therefore 2x^4 u^m \cdot mu^{m-1} \frac{du}{dx} + u^{4m} = 4x^6$$

$$\Rightarrow \frac{du}{dx} = \frac{4x^6 - u^{4m}}{2mx^4 \cdot u^{2m-1}}$$

This equation is homogeneous,

$$\text{if } 4m = 6 \Rightarrow m = 3/2$$

$$\text{Or } 4 + 2m - 1 = 6$$

$$\therefore m = 3/2$$

Q9 (1) - Single Correct

Equation of normal of curve at point $P(x, y)$ is

$$Y - y = -\frac{dx}{dy}(X - x)$$

It passes through $(1, 0)$.

$$a - y = -\frac{dx}{dy}(1 - x)$$

$$ydy = (1 - x)dx$$

On integrating both sides, we get

$$\int ydy = \int (1 - x)dx$$

$$\frac{y^2}{2} = x - \frac{x^2}{2} + C$$

This curve is passes through the origin.

$$\therefore C = 0$$

Hence, equation of curve is

$$x^2 + y^2 - 2x = 0$$

Intersection point of curve $x^2 + y^2 - 2x = 0$ and parabola $y^2 = 4x$ is $(0, 0)$

$\therefore$ Slope of tangent of parabola

$$\text{i.e. } \frac{dy}{dx} = \frac{2}{y}$$

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$$\Rightarrow \left(\frac{dy}{dx} \right)_{(0,0)} = \infty$$

$\therefore$ Equation of common tangent is $x = 0$.

Q10 (2) - Single Correct

We have,

$$\begin{aligned} (x+1)f'(x) - 2(x^2+x)f(x) &= \frac{ex^2}{x+1} \\ \Rightarrow f'(x) - \frac{2x(x+1)}{x+1}f(x) &= \frac{e^{x^2}}{(x+1)^2} \\ \Rightarrow \frac{dy}{dx} - 2xy &= \frac{e^{x^2}}{(x+1)^2} \quad \left[\because y = f(x) \right] \\ &\quad \left[\Rightarrow \frac{dy}{dx} = f'(x) \right] \\ |F = e^{\int -2x dx} &= e^{-x^2} \end{aligned}$$

$\therefore$ Solution of given differential equation is

$$y \cdot e^{-x^2} = \int \frac{e^{x^2}}{(x+1)^2} \cdot e^{-x^2} dx + C$$

$$\Rightarrow ye^{-x^2} = \int \frac{dx}{(x+1)^2} + C$$

$$ye^{-x^2} = \frac{-1}{x+1} + C$$

$$\Rightarrow y = \frac{-e^{x^2}}{x+1} + Ce^{x^2}$$

When $x = 0, y = 5$, then $C = 6$

$$\text{Hence, } y = \frac{-e^{x^2}}{x+1} + 6e^{x^2}$$

$$\Rightarrow y = e^{x^2} \left[\frac{-1+6x+6}{x+1} \right]$$

$$\Rightarrow y = \left(\frac{6x+5}{x+1} \right) e^{x^2}$$

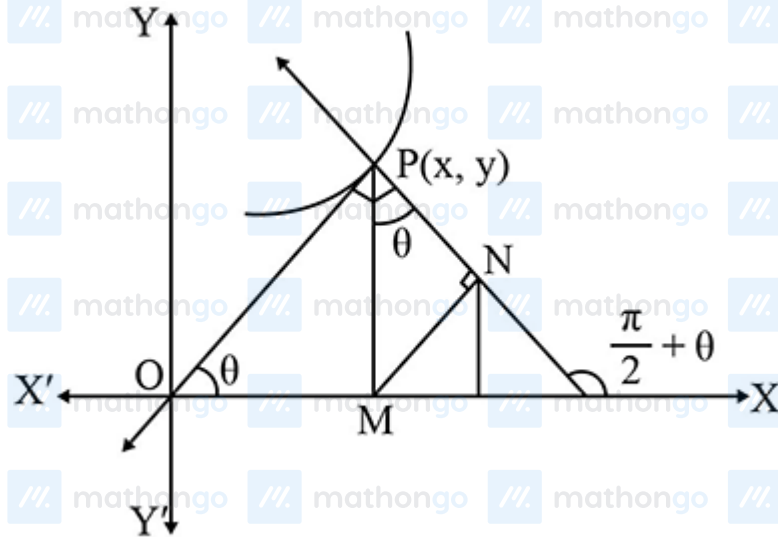
Q11 (1) - Single Correct

Let ordinate = PM

and projection of ordinate on normal = PN

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$$\therefore PN = PM \cos \theta = a$$

$$\Rightarrow y \cos \theta = a$$

$$\Rightarrow y = a \sec \theta$$

$$\Rightarrow y^2 = a^2 (1 + \tan^2 \theta)$$

$$\Rightarrow \tan^2 \theta = \frac{y^2}{a^2} - 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{y^2 - a^2}}{a}$$

$$\Rightarrow [\because \tan \theta = dy/dx]$$

$$\Rightarrow \frac{dx}{a}$$

On integrating both sides, we get

$$a \log |y + \sqrt{y^2 - a^2}| = x + C$$

Q12 (3) - Single Correct

We have, $\frac{dy}{dx} + y \tan x = x \tan x + 1$

$$IF = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

Solution of given differential equation is

$$y \sec x = \int (x \tan x + 1) \sec x dx + C$$

$$\Rightarrow y \sec x = \int (x \sec x \tan x + \sec x) dx + C$$

$$\Rightarrow y \sec x = x \sec x - \int \sec x dx + \int \sec x dx + C$$

$$\Rightarrow y \sec x = x \sec x + C$$

$$\Rightarrow y = x + C \cos x$$

This curve intersect at Y-axis at (0, c)

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$$\Rightarrow \frac{dy}{dx} = 1 - c \sin x$$

$$\therefore \left(\frac{dy}{dx} \right)_{(ac)} = 1$$

Hence, slope is constant for all values of y .

$\therefore$ Tangents are parallel.

Q13 (1) - Single Correct

On differentiating wr.t x , we get

$$\frac{dy}{dx} = \alpha z^{\alpha-1} \frac{dz}{dx}$$

On substituting the value of y and $\frac{dy}{dx}$ in

$$(x^2 y^2 - 1) \frac{dy}{dx} + 2xy^3 = 0$$

$$\Rightarrow (x^2 z^{2\alpha} - 1) \left(\alpha z^{\alpha-1} \frac{dz}{dx} \right) + 2x z^{3\alpha} = 0$$

$$\Rightarrow \frac{dz}{dx} \alpha x^2 z^{3\alpha-1} - \alpha z^{\alpha-1} = 0$$

For homogeneous,

$$3\alpha + 1 = \alpha - 1$$

$$\therefore \alpha = -1$$

Q14 (1) - Single Correct

We have, $\int_0^x ty(t)dt = x^2 y(x)$

On differentiating wr.t. x , we get

$$xy(x) = x^2 y'(x) + 2xy(x)$$

$$\Rightarrow x^2 y'(x) + xy(x) = 0$$

$$\Rightarrow xy'(x) = -y(x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{-y}{x} \quad \left[\because y = y(x) \right]$$

On integrating both sides, we get

$$\int \frac{dy}{y} = - \int \frac{dx}{x}$$

$$\Rightarrow \log y = -\log x + \log C$$

$$\Rightarrow xy = C$$

It passes through $(2, 3)$.

$$\therefore 2 \cdot 3 = C \Rightarrow C = 6$$

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Hence,

$$xy = 6$$

Q15 (1, 2, 4) - Multiple Correct

Equation of normal at any point $P(x, y)$ is

$$Y - y = \frac{-dx}{dy}(X - x)$$

Perpendicular distance from origin,

$$\frac{x + Y \frac{dy}{dx}}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}} = y$$

$$\Rightarrow \left(x + y \frac{dy}{dx}\right)^2 = y^2 \left[1 + \left(\frac{dy}{dx}\right)^2\right]$$

$$\Rightarrow x^2 + 2xy \frac{dy}{dx} + y^2 \left(\frac{dy}{dx}\right)^2 = y^2 + y^2 \left(\frac{dy}{dx}\right)^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy} \text{ [homogenous form]}$$

$$\Rightarrow 2xy \frac{dy}{dx} + x^2 = y^2$$

$$\Rightarrow 2y \frac{dy}{dx} + x = \frac{y^2}{x}$$

$$\Rightarrow 2y \frac{dy}{dx} - \frac{y^2}{x} = -x$$

$$\text{Put } y^2 = z \Rightarrow y \frac{dy}{dx} = \frac{dz}{dx}$$

$$\therefore \frac{dz}{dx} = \frac{z}{x} = -x \Rightarrow \text{Linear differential equation form}$$

$$\text{IF } e^{\int \frac{-1}{x} dx} = e^{-\log x} = \frac{1}{x}$$

Solution of the equation is

$$\frac{z}{x} = \int \frac{-x}{x} dx + C$$

$$\Rightarrow \frac{z}{x} = -x + C$$

$$\Rightarrow z = -x^2 + Cx$$

$$\Rightarrow y^2 = -x^2 + Cx$$

$$\Rightarrow x^2 + y^2 - Cx = 0$$

which is the family of circle touching the Y-axis at origin.

Q16 (3, 4) - Multiple Correct

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We have,

$$f^2(x) + 4f'(x)f(x) + [f'(x)]^2 = 0$$

$$\Rightarrow y^2 + \frac{4dy}{dx} \cdot y + \left(\frac{dy}{dx}\right)^2 = 0 \quad \left[\begin{array}{l} \because y = f(x) \\ \Rightarrow \frac{dy}{dx} = f'(x) \text{ and} \\ y^2 = f^2(x) \end{array} \right]$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 + 4y\frac{dy}{dx} + y^2 = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-4y \pm \sqrt{16y^2 - 4y^2}}{2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-4y \pm 2\sqrt{3}y}{2}$$

$$\Rightarrow \frac{dy}{dx} = -2y \pm \sqrt{3}y$$

$$\Rightarrow \frac{dy}{y} = (-2 \pm \sqrt{3})dx$$

On integrating both sides, we get

$$\log y = (-2 \pm \sqrt{3})x + C_1$$

$$\Rightarrow y = Ce^{(-2 \pm \sqrt{3})x}$$

Q17 (1, 2, 3) - Multiple Correct

We have,

$$\frac{dy}{dx} \sin x - y \cos x + \frac{\sin^2 x}{x^2} = 0$$

$$\Rightarrow \frac{dy}{dx} - y \frac{\cos x}{\sin x} = \frac{-\sin x}{x^2}$$

$$\therefore F = e^{\int \frac{-\cos x}{\sin x} dx} = e^{-\log \sin x} = \frac{1}{\sin x}$$

Solution of the equation is

$$\frac{y}{\sin x} = \int \frac{-\sin x}{x^2} \times \frac{1}{\sin x} dx + C$$

$$\frac{y}{\sin x} = -\int \frac{1}{x^2} dx + C$$

$$\Rightarrow \frac{y}{\sin x} = \frac{1}{x} + C$$

$$\Rightarrow y = \frac{\sin x}{x} + C \sin x$$

When $x \rightarrow \infty$, $y \rightarrow 0$, then $C = 0$

$$\therefore f(x) = \frac{\sin x}{x}$$

$$(1) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$(2) \text{ Let } I = \int_0^{\pi/2} f(x) dx = \int_0^{\pi/2} \frac{\sin x}{x} dx$$

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Since, $\frac{\sin x}{x}$ is decreasing function.

So, $x > 0$

$$\Rightarrow f(x) < f(0)$$

$$\Rightarrow \int_0^{\pi/2} f(x) dx < \frac{\pi}{2} \text{ and } x < \frac{\pi}{2}$$

$$\Rightarrow \int_0^{\pi/2} f(x) dx > 1$$

Q18 (1, 2, 3) - Multiple Correct

We have.

$$f(x) = \int_0^x f(t) \cos t - \cos(t-x) dt$$

$$\Rightarrow f(x) = \int_0^x f(t) \cos t - \int_0^x \cos(t-x) dt$$

$$\Rightarrow f(x) = \int_0^x f(t) \cos t - \int_0^x \cos(-t) dt \left[\because \int_0^a f(x) dx = \int_0^a f(a-x) dx \right]$$

On differentiating both sides, we get

$$f'(x) = f(x) \cos x - \cos x$$

$$\Rightarrow f'(x) - f(x) \cos x = -\cos x$$

$$\Rightarrow \frac{dy}{dx} - y \cos x = -\cos x \left[\because y = f(x) \Rightarrow \frac{dy}{dx} = f'(x) \right]$$

$$IF = e^{\int -\cos x dx} = e^{-\sin x}$$

Solution of the equation is

$$y \cdot e^{-\sin x} = -\int e^{-\sin x} \cos x dx + C$$

$$\Rightarrow y \cdot e^{-\sin x} = e^{-\sin x} + C$$

$$\Rightarrow y = Ce^{\sin x} + 1$$

If $x = 0, y = 0$, then $C = -1$

$$\therefore f(x) = -e^{\sin x} + 1$$

Minimum value of $f(x)$ is $1 - e$ at $x = \pi/2$ and maximum value of $f(x)$ is $1 - e^{-1}$ at $x = -\pi/2$

Now, $f(x) = 1 - e^{\sin x}$

$$\Rightarrow f(x) = -\cos x e^{\sin x}$$

$$\Rightarrow f'(0) = -1$$

$$\Rightarrow f''(x) = -[\cos^2 x e^{\sin x} - \sin x e^{\sin x}]$$

$$f'' \cdot \left(\frac{\pi}{2}\right) = e$$

Q19 (1, 2, 4) - Multiple Correct

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Given, $\frac{dy}{dx} + y = f(x)$

$$IF = e^{\int dx} = e^x$$

$$\Rightarrow ye^x = \int e^x f(x) dx + C$$

Now, if $0 \leq x \leq 2$, then

$$\Rightarrow ye^x = \int e^x e^{-x} dx + C$$

$$\Rightarrow ye^x = x + C$$

When, $x = 0, y = 1$, then $C = 1$

$$\therefore y = xe^{-x} + e^{-x}$$

$$\Rightarrow y(1) = e^{-1} + e^{-1} = 2e^{-1}$$

$$\Rightarrow \frac{dy}{dx} = -xe^{-x} + e^{-x} - e^{-x} = -xe^{-x}$$

Then, $\left(\frac{dy}{dx}\right)_{x=1} = -e^{-1}$.

If $x > 2$ then, $ye^x = \int e^x \cdot e^{-2} dx + C$

$$\Rightarrow ye^x = e^{x-2} + C$$

$$\Rightarrow y = e^{-2} + Ce^{-x}$$

As y is continuous,

$$\lim_{x \rightarrow 2} xe^{-x} + e^{-x} = \lim_{x \rightarrow 2} (e^{-2} + Ce^{-x})$$

$$\Rightarrow 3e^{-2} = (C + 1)e^{-2}$$

$$\Rightarrow C = 2$$

For $x > 2$, $y = e^{-2} + 2e^{-x}$

when $x = 3$

$$y = e^{-2} + 2e^{-3} \Rightarrow y' = -2e^{-x}$$

$$\therefore y'(3) = -2e^{-3}$$

Q20 (1, 3) - Multiple Correct

Given,

$$\frac{dy}{dx} = \frac{ax + h}{by + k}$$

$$\Rightarrow (by + k)dy = (ax + h)dx$$

On integrating both sides, we get

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$$\int (by + k)dy = \int (ax + h)dx$$

$$\Rightarrow \frac{by^2}{2} + ky = \frac{ax^2}{2} + hx + c$$

$$\Rightarrow by^2 + 2ky = ax^2 + 2hx + c$$

It represents parabola of either $a = 0, b \neq 0$ or $a \neq 0, b = 0$.

Q21 (1, 3) - Multiple Correct

The equation of normal at (x, y) is

$$(X - x) + (Y - y)\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{X - x}{x + y\frac{dy}{dx}} + \frac{Y - y}{\frac{x + y\frac{dy}{dx}}{dy/dx}} = 1$$

$$\Rightarrow OA = x + y\frac{dy}{dx} \text{ and } OB = \frac{x + y\frac{dy}{dx}}{dy/dx}$$

$$\text{Also, } \frac{1}{OA} + \frac{1}{OB} = 1$$

$$\Rightarrow 1 + \frac{dy}{dx} = x + y\frac{dy}{dx}$$

$$\Rightarrow (y - 1)dy/dx + (x - 1) = 0$$

On integrating both sides, we get

$$(y - 1)^2 + (x - 1)^2 = C$$

Since, the curve passes through the point $(5, 4)$ and $C = 25$.

$$\text{Hence, the curve is } (x - 1)^2 + (y - 1)^2 = (5)^2.$$

$\therefore$ Centre of circle = $(1, 1)$ and radius = 5.

Q22 (1, 2, 4) - Multiple Correct

$$\text{Given, } \frac{dy}{dx} + y \cos x = \cos x$$

$$\Rightarrow \frac{dy}{dx} = \cos x (y - 1)$$

$$\Rightarrow \frac{dy}{y-1} = \cos x dx$$

On integrating both sides, we get

$$\int \frac{dy}{y-1} = \int \cos x dx$$

$$\Rightarrow \log(y - 1) = \sin x + C$$

$$\Rightarrow y - 1 = Ce^{\sin x}$$

Since, it passes through the point $(0, 1)$.

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$$\therefore 1 - 1 = Ce^0$$

$$\Rightarrow C = 0$$

$$\Rightarrow y = 1$$

$$\therefore f(x) = 1$$

So, it is constant function and it is continuous and differentiable, $\forall x \in R$

Hence, every constant function is periodic.

Q23 (1, 2) - Multiple Correct

We have, $y' + y = \cos x - \sin x$

$$\Rightarrow \frac{dy}{dx} + y = \cos x - \sin x$$

$$\therefore IF = e^{\int dx} = e^x$$

$\therefore$ Hence, the solution of given differential equation is

$$y \cdot e^x = \int e^x (\cos x - \sin x) dx + C$$

$$\Rightarrow y \cdot e^x = e^x \cos x + C$$

$$\Rightarrow y = \cos x + Ce^{-x}$$

When $x = \frac{\pi}{2}$ and $y = 0$, then $C = 0$

$$\therefore y(0) = 1, y\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \text{ and } y\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Q24 (1, 3) - Multiple Correct

We have, $y' + y \tan x = 2x + x^2 \tan x$

$$\Rightarrow \frac{dy}{dx} + y \tan x = 2x + x^2 \tan x$$

$$\therefore IF = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

Hence, the solution of given differential equation is

$$y \sec x = \int (2x + x^2 \tan x) \sec x dx + C$$

$$\Rightarrow y \sec x = \int (2x \sec x + x^2 \sec x \tan x) dx + C$$

$$\Rightarrow y \sec x = \sec x \int 2x dx - \int \frac{d}{dx} \sec x \int 2x dx \cdot dx + \int x^2 \sec x \tan x dx + C$$

$$\Rightarrow y \sec x = x^2 \sec x - \int x^2 \sec x \tan x dx + \int x^2 \sec x \tan x dx + C$$

$$\Rightarrow y \sec x = x^2 \sec x + C$$

When $x = 0$ and $y = 0$, then $C = 0$

$$\therefore y = x^2$$

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$$\text{Now, } y\left(\frac{\pi}{4}\right) = \left(\frac{\pi}{4}\right)^2 = \frac{\pi^2}{16}$$

$$\text{and } y\left(\frac{\pi}{3}\right) = \left(\frac{\pi}{3}\right)^2 = \frac{\pi^2}{9}$$

Q25 (2, 3) - Multiple Correct

$$\text{We have, } y' + \frac{y}{x} = \cos x + \frac{\sin x}{x}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \cos x + \frac{\sin x}{x}$$

$$\text{IF} = e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

Hence, the solution of given differential equation is

$$y \cdot x = \int \left(\cos x + \frac{\sin x}{x} \right) x dx + C$$

$$\Rightarrow y \cdot x = \int (x \cos x + \sin x) dx + C$$

$$\Rightarrow y \cdot x = x \sin x + C$$

$$\Rightarrow y = \sin x + C/x$$

When $x = 0$ and $y = 0$, then $C = 0$

$$\therefore y = \sin x$$

$$\text{Now, } y\left(\frac{\pi}{2}\right) = 1 \text{ and } y\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

Q26 (1, 3) - Multiple Correct

$$\text{We have, } y - (x + 2y^2) y' = 0$$

$$\Rightarrow y - (x + 2y^2) \frac{dy}{dx} = 0 \Rightarrow \frac{dx}{dy} = \frac{x + 2y^2}{y}$$

$$\Rightarrow \frac{dx}{dy} - \frac{x}{y} = 2y$$

$$\therefore IF = e^{\int \frac{-1}{y} dy} = e^{-\log y} = \frac{1}{y}$$

Hence, the solution of given differential equation is

$$\frac{x}{y} = \int \frac{2y}{y} dy + C \Rightarrow \frac{x}{y} = 2y + C$$

$$\Rightarrow x = 2y^2 + Cy$$

When $x = 0$, $y = 1$, then $C = -2$

$$\therefore x = 2y^2 - 2y \Rightarrow x = 2y(y - 1)$$

When $y = 2$, then $x = 4$

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Q27 (1) - Multiple Correct

We have, $\frac{xdx+yd y}{x d y-y d x}=\sqrt{\frac{1-x^2-y^2}{x^2+y^2}}$

$$\Rightarrow \frac{xdx+yd y}{\sqrt{1-(x^2+y^2)}}=\frac{xd y-y d x}{\sqrt{x^2+y^2}}$$

$$\Rightarrow \frac{xdx+yd y}{(\sqrt{x^2+y^2}) \sqrt{1-(x^2+y^2)}}=\frac{xd y-y d x}{x^2+y^2}$$

$$\Rightarrow d\left(\sin ^{-1} \sqrt{x^2+y^2}\right)=d \tan ^{-1}\left(\frac{y}{x}\right)$$

$$\Rightarrow \int d\left(\sin ^{-1} \sqrt{x^2+y^2}\right)=\int d \tan ^{-1}\left(\frac{y}{x}\right)$$

$$\Rightarrow \sin ^{-1} \sqrt{x^2+y^2}=\tan ^{-1}\left(\frac{y}{x}\right)+C$$

When $x=1, y=0$, then $C=\frac{\pi}{2}$

$$\therefore \sin ^{-1} \sqrt{x^2+y^2}=\tan ^{-1}\left(\frac{y}{x}\right)+\frac{\pi}{2}$$

when $x=0, y=0$

Q28 (3, 4) - Multiple Correct

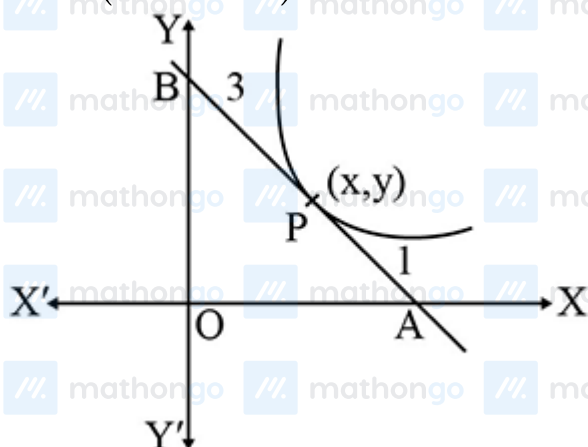
Equation of tangent to the curve $y=f(x)$ at (x, y) is

$$Y-y=\frac{d y}{d x}(x-x)$$

Thus, cuts the X -axis at A and Y -axis at B .

$$\therefore A\left(\frac{x \frac{d y}{d x}-y}{\frac{d y}{d x}}, 0\right)$$

and $B\left(0,-x \frac{d y}{d x}+y\right)$



Now, it is given that $BP:AP=3:1$

$$x=\frac{3\left(\frac{x \frac{d y}{d x}-y}{\frac{d y}{d x}}\right)+1 \times 0}{3+1}$$

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$$\Rightarrow 4 \times \frac{dy}{dx} = 3 \times \frac{dy}{dx} - 3y \Rightarrow x \frac{dy}{dx} + 3y = 0$$

$$\int \frac{dy}{y} = \int \frac{-3dx}{x}$$

$$\Rightarrow \log y = -3 \log x + \log C$$

$$\Rightarrow yx^3 = C$$

Curve passes through (1, 1)

$$\therefore C = 1$$

$$\text{Equation of curve} = x^3 y = 1$$

The curve is also passes through $\left(2, \frac{1}{8}\right)$.

Q29 (2) - Paragraph 1

$$\text{Given, } f(x) = (e^{-x} + e^x) \cos x - 2x - \int_0^x (x-t)f'(t)dt$$

$$f(0) = (1+1) - 0 - 0 = 2$$

$$\therefore f(0) = 2$$

$$\text{Now, } f(x) = (e^{-x} + e^x) \cos x - 2x - x \int_0^x f'(t)dt + \int_0^x t f'(t)dt$$

$$\Rightarrow f(x) = (e^{-x} + e^x) \cos x - 2x - x[f(x)]_0^x + \int_0^x t f'(t)dt$$

$$\Rightarrow f(x) = (e^{-x} + e^x) \cos x - 2x - x f(x) + x f(0) + \int_0^x t f'(t)dt$$

$$\Rightarrow f(x) = (e^{-x} + e^x) \cos x - 2x - x f(x) + 2x + \int_0^x t f'(t)dt$$

$$\Rightarrow f(x) = (e^{-x} + e^x) \cos x - x f(x) + \int_0^x t f'(t)dt$$

On differentiating w.r.t. x , we get

$$f'(x) = (e^{-x} + e^x)(-\sin x) + \cos x(-e^{-x} + e^x) - x f'(x) - f(x) + x f'(x)$$

$$\Rightarrow f'(x) = e^x(\cos x - \sin x) - e^{-x}(\cos x + \sin x) - f(x)$$

$$\Rightarrow f'(x) + f(x) = e^x(\cos x - \sin x) - e^{-x}(\cos x + \sin x)$$

$$\Rightarrow \frac{dy}{dx} + y = e^x(\cos x - \sin x) - e^{-x}(\cos x + \sin x)$$

Q30 (4) - Paragraph 1

$$f'(x) + f(x) = e^x(\cos x - \sin x) - e^{-x}(\cos x + \sin x)$$

$$\Rightarrow f''(x) + f'(x) = e^x[\cos x - \sin x - \sin x - \cos x] - e^{-x}[-\cos x - \sin x - \sin x + \cos x]$$

$$\Rightarrow f''(x) + f'(x) = -2e^x \sin x + 2e^{-x} \sin x$$

$$\Rightarrow f''(0) + f'(0) = -2 + 2 = 0$$

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Q31 (1) - Paragraph 2

Given, $(1 + x^2) \frac{dy}{dx} + 2xy = 4x^2$

$$\Rightarrow \frac{dy}{dx} + \frac{2xy}{1+x^2} = \frac{4x^2}{1+x^2}$$

$$\therefore IF = e^{\int \frac{2x}{1+x^2} dx} = e^{\log|1+x^2|} = 1 + x^2$$

Solution of the equation is

$$y(1 + x^2) = \int \frac{4x^2}{1+x^2} \times (1 + x^2) dx$$

$$y(1 + x^2) = \int 4x^2 dx$$

$$y(1 + x^2) = \frac{4x^3}{3} + C$$

when, $x = 0, y = 0$, then $C = 0$

$$\therefore y = \frac{4x^3}{3(1+x^2)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{4}{3} \left[\frac{(1+x^2)(3x^2) - x^3(2x)}{(1+x^2)^2} \right] \Rightarrow \frac{dy}{dx} = \frac{4}{3} \left[\frac{3x^2 + x^4}{(1+x^2)^2} \right]$$

$$\frac{dy}{dx} > 0, \forall x \neq 0$$

$\frac{dy}{dx} = 0$ at $x = 0$ and it does not change sign.

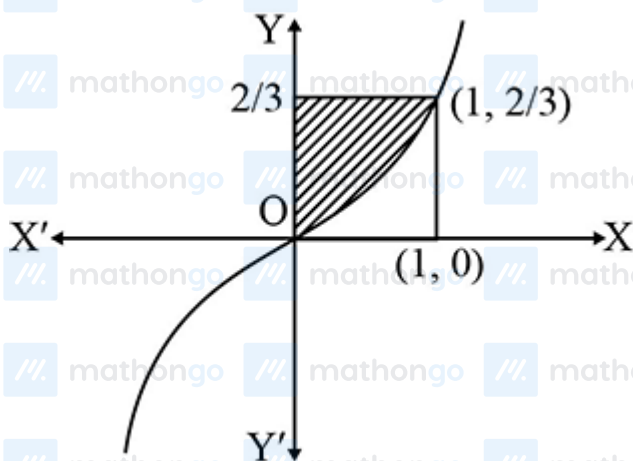
$\Rightarrow x = 0$ is the point of inflection.

$\therefore y = f(x)$ is increasing, $\forall x \in R$.

Q32 (3) - Paragraph 2

We have, $y = \frac{4}{3} \left(\frac{x^3}{1+x^2} \right)$

Since, $x \rightarrow \infty, y \rightarrow \infty$ and $x \rightarrow -\infty, y \rightarrow -\infty$



Area enclosed by $y = f^{-1}(x)$

X -axis and ordinate at $x = 2/3$

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$$A = \frac{2}{3} - \frac{4}{3} \int_0^1 \frac{x^3}{1+x^2} dx$$

$$\text{Put } 1+x^2 = t \Rightarrow 2x dx = dt$$

$$\Rightarrow A = \frac{2}{3} - \frac{2}{3} \int_1^2 \frac{t-1}{t} dt$$

$$\Rightarrow A = \frac{2}{3} - \frac{2}{3} [t - \log t]_1^2$$

$$\therefore A = \frac{2}{3} [1 - 2 + \log 2 + 1] = \frac{2}{3} \log_e 2$$

Q33 (2) - Paragraph 3

$$\text{Given, } (1+y^2) dx - xy dy = 0$$

$$\Rightarrow \frac{y}{1+y^2} dy = \frac{dx}{x}$$

On integrating both sides, we get

$$\int \frac{y}{1+y^2} dy = \int \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \log(1+y^2) = \log x C$$

$$\Rightarrow 1+y^2 = x^2 C^2$$

Since, it passes through the point (1, 0).

$$C^2 = 1$$

Hence, the equation of the curve is

$$x^2 - y^2 = 1$$

Q34 (1) - Paragraph 3

$$\text{Given, } \frac{dy}{dx} + \frac{1+y^2}{\sqrt{1-x^2}} = 0$$

$$\Rightarrow \frac{dy}{y^2+1} = \frac{-dx}{\sqrt{1-x^2}}$$

$$\Rightarrow \int \frac{dy}{y^2+1} = - \int \frac{dx}{\sqrt{1-x^2}}$$

$$\Rightarrow \tan^{-1} y = -\sin^{-1} x + C$$

$$\therefore \tan^{-1} y + \sin^{-1} x = C$$

Q35 (3) - Paragraph 4

$$\text{We have, } \frac{dy}{dx} = \frac{y}{x} - \cot\left(\frac{y}{x}\right) \cos\left(\frac{y}{x}\right)$$

$$\Rightarrow v + x \frac{dv}{dx} = v - \cot v \cos v \left[\because y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx} \right]$$

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$$\Rightarrow x \frac{dv}{dx} = -\frac{\cos^2 v}{\sin v}$$

$$\Rightarrow \frac{\sin v}{\cos^2 v} dv = -\frac{dx}{x}$$

On integrating both sides, we get

$$\int \sec v \tan v dv = -\int \frac{dx}{x}$$

$$\Rightarrow \sec v = -\log x + C$$

$$\Rightarrow \sec\left(\frac{y}{x}\right) = \log \frac{1}{x} + C$$

$$\text{Hence, } f(x) = \log |1/x|$$

Q36 (1) - Paragraph 4

$$\begin{aligned} \text{Now, } \lim_{x \rightarrow 0} \frac{f(1+x)}{x} &= \lim_{x \rightarrow 0} \frac{\log \frac{1}{1+x}}{x} \left[\because f(x) = \log\left(\frac{1}{x}\right) \right] \\ &= \lim_{x \rightarrow 0} \frac{-\log(1+x)}{x} = -1 \end{aligned}$$

Q37 (3) - Paragraph 5

$$\text{Given, } f''(x) - 2f'(x) + f(x) \geq e^x$$

$$\Rightarrow f''(x)e^{-x} - 2f'(x)e^{-x} + f(x)e^{-x} \geq 1$$

$$\Rightarrow f''(x)e^{-x} - f'(x)e^{-x} - f'(x)e^{-x} + f(x)e^{-x} \geq 1$$

$$\Rightarrow \frac{d[f'(x)e^{-x}]}{dx} - \frac{d[f(x)e^{-x}]}{dx} \geq 1$$

$$\Rightarrow \frac{d^2}{dx^2} [e^{-x} f(x)] \geq 1, \forall x \in [0, 1]$$

$\therefore \phi(x) = e^{-x} f(x)$ is concave up

$$f(0) = f(1) = 0 \Rightarrow \phi(0) = 0 = \phi(1)$$

$$\text{Here, } \phi'(x) < 0, x \in \left(0, \frac{1}{4}\right) \text{ and } \phi'(x) > 0, x \in \left(\frac{1}{4}, 1\right)$$

$$\Rightarrow e^{-x} f'(x) - e^{-x} f(x) < 0, x \in \left(0, \frac{1}{4}\right)$$

$$\Rightarrow f'(x) < f(x), x \in \left(0, \frac{1}{4}\right)$$

Q38 (4) - Paragraph 5

$$\text{Here, } \phi(x) < 0 \Rightarrow e^{-x} f(x) < 0$$

$$\therefore f(x) < 0$$

$$\text{Hence, } -\infty < f(x) < 0$$

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Q39 (4) - Integer Type

Given, $y' + \frac{y}{x} = \frac{1}{(1+\log xy)^2}$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{1}{(1+\log xy)^2} \dots(i)$$

Let $xy = v$

$$\Rightarrow x \frac{dy}{dx} + y = \frac{dv}{dx} \Rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{1}{x} \frac{dv}{dx}$$

Eq. (i) reduces to

$$\frac{1}{x} \frac{dv}{dx} = \frac{1}{(1+\log v)^2}$$

$$\Rightarrow (1 + \log v)^2 dv = x dx$$

On integrating, we get

$$\int (1 + 2 \log v + (\log v)^2) dv = \int x dx$$

$$\Rightarrow v + \int (\log v)^2 dv + 2 \int \log v dv = \frac{x^2}{2} + C$$

$$\Rightarrow v + v(\log v)^2 - \int \frac{2 \log v}{v} \cdot v dv + 2 \int \log v dv = \frac{x^2}{2} + C$$

$$\Rightarrow v + v(\log v)^2 = \frac{x^2}{2} + C$$

$$\Rightarrow v[1 + (\log v)^2] = \frac{x^2}{2} + C$$

$$\Rightarrow xy [1 + (\log xy)^2] = \frac{x^2}{2} + C$$

$$[\because v = xy]$$

$$\lambda = 2, \mu = 2 \Rightarrow \lambda + \mu = 4$$

Q40 (5) - Integer Type

Given, $y' (x^2 - 2x^3y^3) = (3x^2y^4 + 2xy)$

$$\Rightarrow dy (x^2 - 2x^3y^3) = (3x^2y^4 + 2xy) dx$$

$$\Rightarrow x^2 dy - 2x^3 y^3 dy - 3x^2 y^4 dx - 2xy dx = 0$$

$$\Rightarrow 3x^2 y^4 dx + 2x^3 y^3 dy + 2xy dx - x^2 dy = 0$$

$$\Rightarrow 3x^2 y^2 dx + 2x^3 y dy + \frac{2xy dx - x^2 dy}{y^2} = 0$$

$$\Rightarrow d(x^3 y^2) + d\left(\frac{x^2}{y}\right) = 0$$

On integrating both sides, we get

$$x^3 y^2 + \frac{x^2}{y} = C$$

$$\lambda = 3, \mu = 2$$

$$\Rightarrow \lambda + \mu = 3 + 2 = 5$$

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Q41 (2) - Integer Type

Given, $y(2xy + 1)dx + x(1 + 2xy + x^2y^2)dy = 0$

$$\Rightarrow 2xy^2dx + ydx + xdy + 2x^2ydy + x^3y^2dy = 0$$

$$\Rightarrow 2xy^2dx + ydx + xdy + 2x^2ydy = -x^3y^2dy$$

$$\Rightarrow \frac{(2xy+1)(ydx+xdy)}{x^3y^3} = \frac{-1}{y}dy$$

$$\Rightarrow \left(2 + \frac{1}{xy}\right) \left(\frac{ydx+xdy}{x^2y^2}\right) = \frac{-1}{y}dy$$

$$\Rightarrow -\int \left(2 + \frac{1}{xy}\right) d\left(\frac{1}{xy}\right) = \frac{-1}{y}dy$$

On integrating both sides, we get

$$\frac{-2}{xy} - \frac{1}{2(xy)^2} = -\log y + C$$

$$\Rightarrow \frac{-2}{xy} - \frac{1}{2(xy)^2} + \log y = C$$

Hence, $\lambda = 2$

Q42 (4) - Integer Type

Given $\cos\left(\frac{x}{y}\right)(ydx - xdy) = xy^3(xdy + ydx)$

$$\cos\left(\frac{x}{y}\right) \left(\frac{ydx - xdy}{y^2}\right) = xy(xdy + ydx)$$

$$\Rightarrow \cos\left(\frac{x}{y}\right) d\left(\frac{x}{y}\right) = xy d(xy)$$

On integrating, we get

$$\sin\left(\frac{x}{y}\right) = \frac{(xy)^2}{2} + C$$

$$\therefore \lambda = 2, \quad \mu = 2$$

Hence, $\lambda + \mu = 2 + 2 = 4$

Q43 (1) - Integer Type

Given, $\left\{y\left(1 + \frac{1}{x}\right) + \cos y\right\}dx + \{x + \log x - x \sin y\}dy = 0$

$$\Rightarrow ydx + \frac{y}{x}dx + \cos ydx + xdy + \log xdy - x \sin ydy = 0$$

$$\Rightarrow (ydx + xdy) + \frac{y}{x}dx + \log xdy + \cos ydx - x \sin ydy = 0$$

$$\Rightarrow dxy + d(y \log x) + d(x \cos y) = C$$

On integrating both sides, we get

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$$xy + y \log x + x \cos y = 0$$

$$\therefore \lambda = 1$$

Q44 (0) - Integer Type

We have,

$$xy' - y = mx^2 \sqrt{x^2 - y^2}$$

$$\Rightarrow xdy - ydx = mx^2 \sqrt{x^2 - y^2} dx$$

$$\Rightarrow \frac{xdy - ydx}{\sqrt{x^2 - y^2}} = mx^2 dx$$

$$\Rightarrow \frac{xdy - ydx}{x \sqrt{1 - \frac{y^2}{x^2}}} = mx^2 dx$$

$$\Rightarrow \frac{xdy - ydx}{\sqrt{1 - y^2/x^2}} = mxdx$$

$$\Rightarrow d \sin^{-1} \left(\frac{y}{x} \right) = mxdx$$

On integrating both sides, we get

$$\sin^{-1} \frac{y}{x} = \frac{mx^2}{2} + C$$

$$\Rightarrow \lambda = 2, \mu = 2$$

$$\Rightarrow \lambda - \mu = 2 - 2 = 0$$

Q45 (1) - Integer Type

$$\text{Given, } \frac{dy}{dx} = (x + y - 1) + \frac{x+y}{\log(x+y)}$$

$$\text{Put } x + y = v \Rightarrow \frac{dy}{dx} = \frac{dv}{dx} - 1$$

$$\frac{dv}{dx} - 1 = (v - 1) + \frac{v}{\log v}$$

$$\Rightarrow \frac{dv}{dx} = \frac{v \log v + v}{\log v}$$

$$\Rightarrow \frac{v \log v + v}{v(\log v + 1)} dv = dx$$

On integrating both sides, we get

$$\int \frac{\log v}{v(\log v + 1)} dv = \int dx$$

$$\Rightarrow \int \left[\frac{1}{v} - \frac{1}{v(\log v + 1)} \right] dv = \int dx$$

$$\Rightarrow \log v - \log(\log v + 1) = x + C$$

$$\Rightarrow \log(x + y) - \log[\log(x + y) + 1] = x + C$$

$$\therefore k = 1$$

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Q46 (9) - Integer Type

Let equation of tangent to the curve $y = f(x)$ at (x, y) is

$$Y - y = \frac{dy}{dx}(X - x)$$

Since, y -intercept of tangent is $-\frac{xdy}{dx} + y$.

It is given that, $-\frac{xdy}{dx} + y = x^3$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = -x^2$$

$$\Rightarrow \frac{xdy - ydx}{x^2} = -x dx$$

On integrating both sides, we get

$$\int d\left(\frac{y}{x}\right) = -\int x dx$$

$$\Rightarrow \frac{y}{x} = -\frac{x^2}{2} + C$$

$$\Rightarrow y = -\frac{x^3}{2} + Cx$$

Since, it passes through the point $(1, 1)$.

$$\therefore 1 = -\frac{1}{2} + C$$

$$\Rightarrow C = \frac{3}{2}$$

When $x = -3$, then

$$y = \frac{-(-3)^3}{2} + \left(\frac{-9}{2}\right)$$

$$= \frac{27}{2} - \frac{9}{2}$$

$$= \frac{18}{2} = 9$$

Q47 (2) - Integer Type

We have,

$$\frac{dy}{dx} = \frac{\sin y + x}{\sin 2y - x \cos y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin y + x}{2 \sin y \cos y - x \cos y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin y + x}{\cos y(2 \sin y - x)}$$

$$\Rightarrow 2 \sin y \cos y dy - x \cos y dy = \sin y dx + x dx$$

$$\Rightarrow 2 \sin y \cos y dy = \sin y dx + x \cos y dy + x dx$$

On integrating both sides, we get

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$$\sin^2 y = x \sin y + \frac{x^2}{2} + C$$

$$\text{Hence, } a = 2$$

Q48 (1) - Integer Type

We have,

$$(1 + 2e^{x/y}) dx + 2e^{x/y}(1 - x/y)dy = 0$$

$$\Rightarrow \frac{dx}{dy} = \frac{-2e^{x/y}(1 - x/y)}{1 + 2e^{x/y}}$$

$$\Rightarrow v + y \frac{dv}{dy} = -\frac{2e^v(1-v)}{1+2e^v} \left[\because x = vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy} \right]$$

$$\Rightarrow y \frac{dv}{dy} = \frac{-2e^v + 2ve^v - v - 2ve^v}{1+2e^v}$$

$$\Rightarrow y \frac{dv}{dy} = \frac{-(v+2e^v)}{1+2e^v}$$

$$\Rightarrow \frac{1+2e^v}{v+2e^v} dv = -\frac{dy}{y}$$

On integrating both sides, we get

$$\int \frac{1+2e^v}{v+2e^v} dv = -\int \frac{dy}{y}$$

$$\Rightarrow \log|v + 2e^v| = \log C/y$$

$$\Rightarrow v + 2e^v = C/y$$

$$\Rightarrow \frac{x}{y} + 2e^{x/y} = \frac{C}{y}$$

$$\Rightarrow x + 2ye^{x/y} = C$$

$$\text{Hence, } f(x) = e^x$$

$$\therefore \lim_{x \rightarrow 0} \frac{f(x)-1}{x} = \lim_{x \rightarrow 0} \frac{e^x-1}{x} = 1$$

Q49 (6) - Integer Type

$$\text{Given, } 6 \int_1^x f(x) dx = 3xf(x) - x^3$$

On differentiating both sides w.r.t. x , we get

$$6f(x) = 3xf'(x) + 3f(x) - 3x^2$$

$$\Rightarrow 2f(x) = xf'(x) + f(x) - x^2 \Rightarrow xf'(x) - f(x) = x^2$$

$$\Rightarrow f'(x) - \frac{f(x)}{x} = x \Rightarrow \frac{dy}{dx} - \frac{y}{x} = x \left[\because y = f(x) \Rightarrow \frac{dy}{dx} = f'(x) \right]$$

which is a form of linear differential equation. where, $P = \frac{-1}{x}$ and $Q = x$

$$\therefore IF = e^{\int \frac{-1}{x} dx} = e^{-\log x} = \frac{1}{x}$$

Hence, the solution of given differential equation is

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$$\frac{y}{x} = \int \frac{x}{x} dx + C$$

$$\frac{y}{x} = x + C \Rightarrow y = x^2 + Cx$$

When $x = 1$ and $y = 2$, then $C = 1$

$$\therefore y = x^2 + x$$

When $x = 2$, then $y = 6$

Q50 (1) - Integer Type

$$\text{Given, } \int_0^1 f(tx) dt = n \cdot f(x)$$

$$\text{Put } tx = u \Rightarrow \frac{dt}{du} = \frac{1}{x}$$

Form Eq. (i) reduces to

$$\frac{1}{x} \int_0^x f(u) du = n f(x)$$

$$\Rightarrow \int_0^x f(u) du = n x f(x)$$

On differentiating w.r.t. x both sides, we get

$$f(x) = n f(x) + n x f'(x)$$

$$\Rightarrow n x f'(x) + (n - 1) f(x) = 0$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{1 - n}{n x}$$

On integrating both sides, we get

$$\log f(x) = \frac{1-n}{n} \log x + \log C$$

$$\Rightarrow \log f(x) = \log x^{\frac{1-n}{n}} \cdot C$$

$$\Rightarrow f(x) = C x^{\frac{1-n}{n}}$$

Hence, $\lambda = 1 - n$ and $\mu = n$

$$\Rightarrow \lambda + \mu = 1 - n + n = 1$$